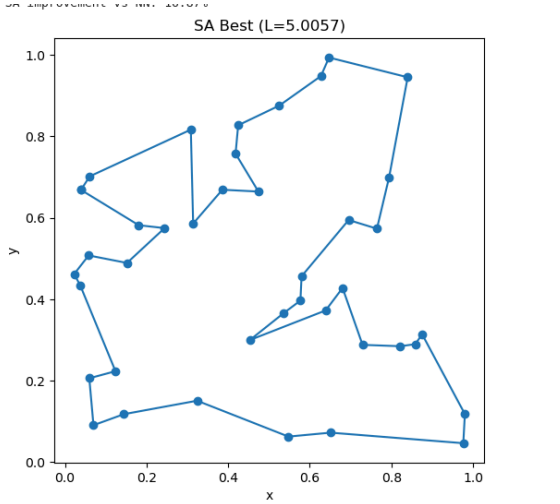
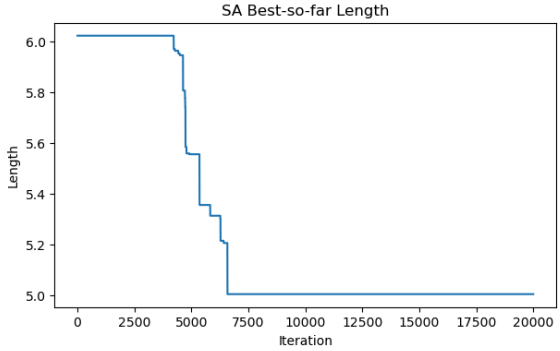
Here, the problem is to find the shortest path between a number of randomly distributed cities. I was given code that gives me an imperfect solution, and asked to create optimization algorithms to improve it. The first one was simulated annealing, which works by randomly switching stuff around, accepting solutions that are slightly worse in the short term in order to escape local minima, and gradually tightening up its restrictions.
here is the code that decides whether a solution is good enough depending on the temperature:

```
accept = False

    if delta <= 0:
        accept = True
    elif delta > 0:
        threshold = rng.random()
        probability = math.exp(-delta / T);
        if threshold <= probability:
            accept = True
```

here are the visualizations the template created:





I was also asked to make a genetic algorithm that simulates evolution with mutations and etc to try and find a good fit:

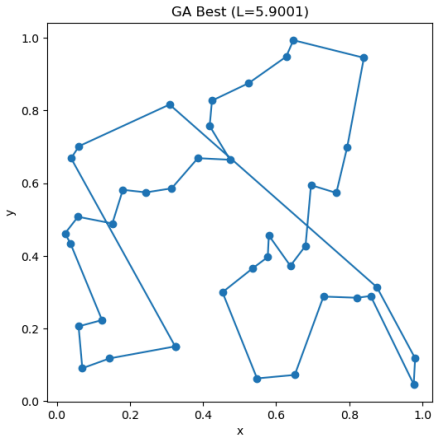
```
def order_crossover_ox(parent1: Tour, parent2: Tour, rng: random.Random) -> Tour:
    n = len(parent1)
    a, b = sorted(rng.sample(range(n), 2))
    child = [None] * n
    used = []
    for i in range(a, b + 1):
        child[i] = parent1[i]
        used.append(parent1[i])
    child_index = (b + 1) % n
```

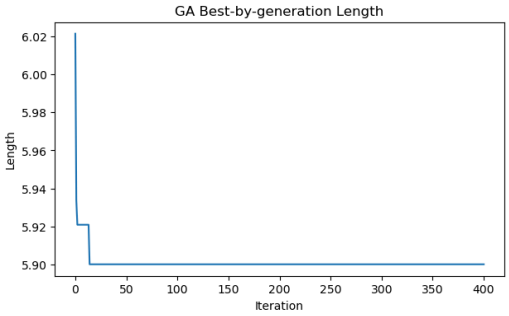
```
    for offset in range(n):
        num = parent2[(b + 1 + offset) % n]
        if num in used:
            continue
        while child[child_index] is not None:
            child_index = (child_index + 1) % n
        child[child_index] = num
        used.append(num)
    return child
```

That is a function I wrote as part of it, which serves to combine 2 solutions together.

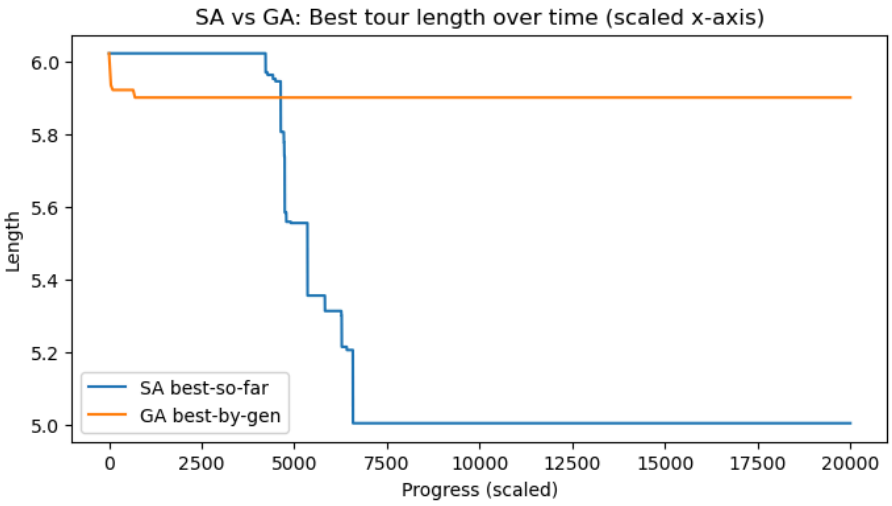
```
def mutate_swap(tour: Tour, rng: random.Random) -> Tour:
    mutated = list(tour)
    i, j = rng.sample(range(len(mutated)), 2)
    mutated[i], mutated[j] = mutated[j], mutated[i]
    return mutated
```

there is also this one, which swaps 2 random cities in the problem.
all in all, the genetic algorithm produced these graphs:





Here is a comparison of the two graphs:



GA converged much faster, but settled into a local minimum really fast for reasons I am not quite sure of. Meanwhile, SA didn't seem to get stuck in that local minimum and reached a much better solution after a much longer amount of time.